Glioma Grade Classification

Comparing four machine learning models

Can clinical and mutation features predict the dataset's LGG or GBM label?

839	23	4	30
original records	input features	models	shared splits

An educational study, not a medical diagnostic system.

What are we trying to distinguish?

CLASS 0

Lower-Grade Glioma

LGG

CLASS 1

Glioblastoma

GBM

All records are from people with glioma. We are predicting grade labels, not deciding whether someone has a brain tumor.

GBM is the positive class for precision, recall, and F1.

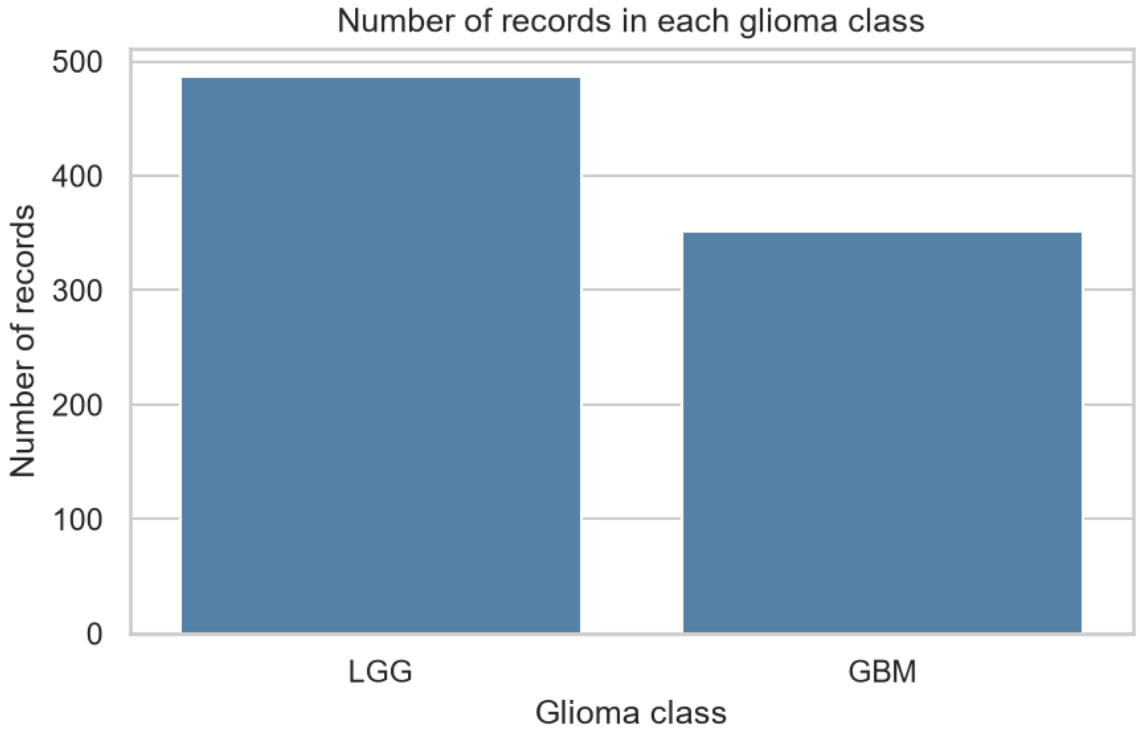
The approved UCI dataset

839 original records

20 gene-mutation features

3 clinical features: age, gender, race

487 LGG and 352 GBM records



Tasci and colleagues (2022) | UCI dataset 759

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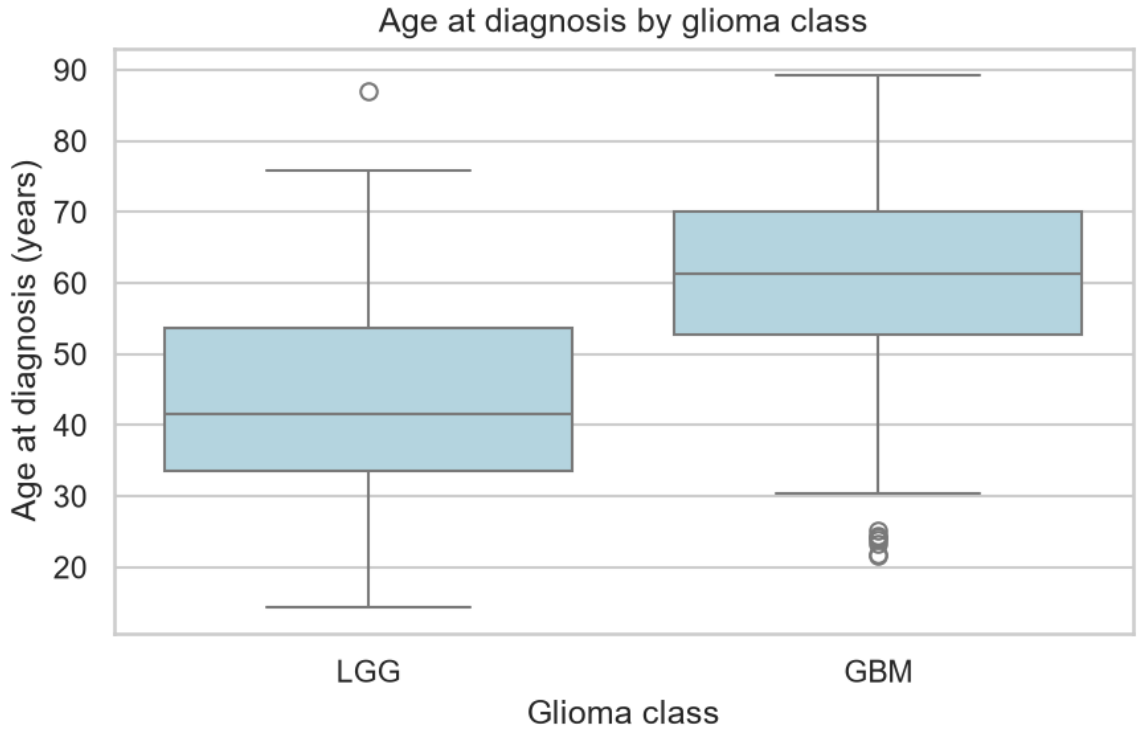
Explore first, then prepare the data

No missing values to fill

Remove one exact repeat: 838 rows

Keep all ages: no IQR flags

Encode Race with yes/no columns



GBM records tend to be older here, but age ranges overlap.

A pattern in this sample does not establish cause and effect.

Four models, the same fair test

Logistic Regression

Random Forest

AdaBoost

XGBoost

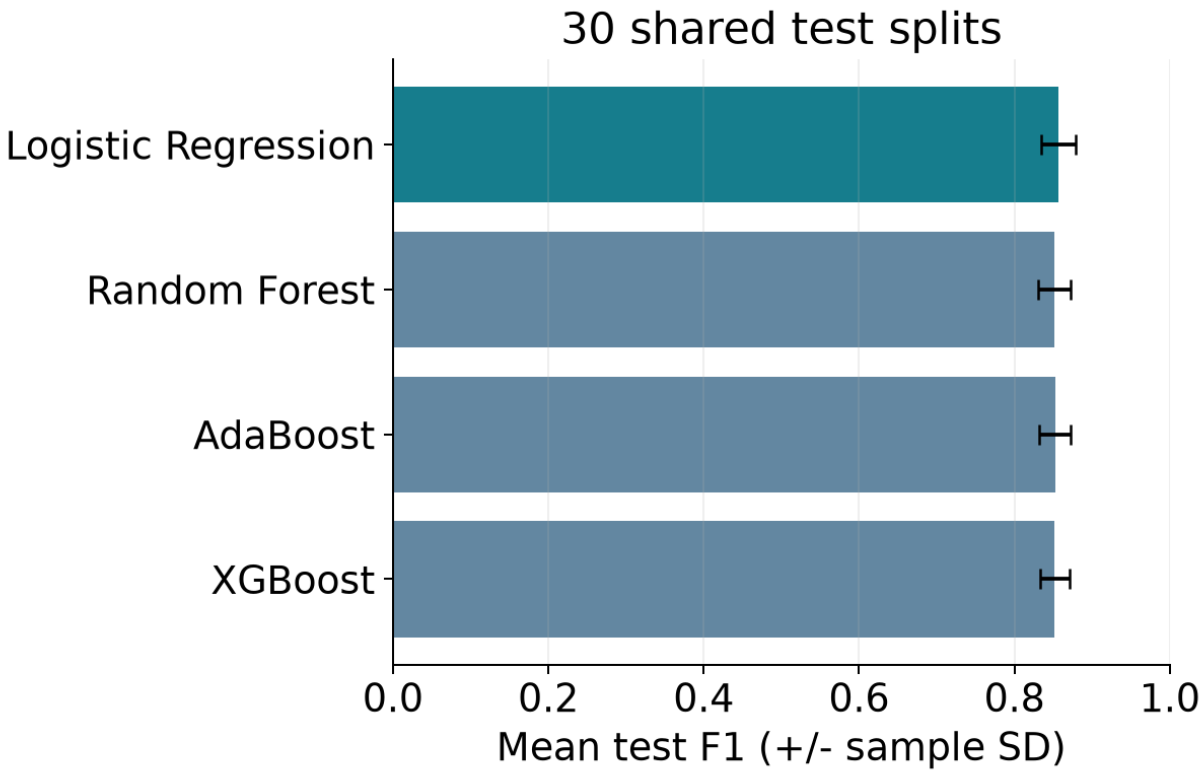
1. Split	80% train / 20% test; same people for all models
2. Tune	Small grids; five-fold CV on training data only
3. Evaluate	Held-out test accuracy, precision, recall, and F1

Repeat the complete process 30 times.

Scale Logistic Regression inside its pipeline; GBM F1 selects settings.

Results: similar scores, a small lead

Model	Accuracy	F1
Logistic Regression	0.8706	0.8566
Random Forest	0.8655	0.8514
AdaBoost	0.8633	0.8523
XGBoost	0.8641	0.8517



Mean test scores across 30 splits.

Clinical features improved mean F1

for only 1 of the 4 models.

Logistic Regression leads mean F1 by just 0.0043. This is not proof of a universal winner.

What we can conclude

A simple model was competitive with the tree ensembles.

Small sample

838 prepared records;

some groups are very small

Overlapping tests

30 splits are not

30 independent studies

No external test

New data and settings

could change the ranking

More testing, privacy protection, and human review are needed

before any practical medical use.